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Virtual Coupling-Enabled Trajectory Optimization for Heavy-Haul Train Group: A Goal-Oriented MADDPG Approach

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***Abstract*—The virtual coupling-based train group operation presents promising prospects to enhance the capacity of freight railways. Adopting a leader-follower control mechanism enables coordinated multi-train operations. However, inhomogeneous environmental disturbances demand-optimized group-level operational states and trajectories during emergencies, particularly on heavy-haul lines. This paper addresses this challenge by developing a mixed-integer quadratic programming (MIQP) model that incorporates safety constraints, train dynamics, speed limits, and tracking interval requirements. A goal-oriented reinforcement learning method based on Multi-Agent Deep Deterministic Policy Gradient (GO-MADDPG) is proposed to achieve the train group's safe, efficient, energy-saving, and stable trajectories. Agents output continuous acceleration and deceleration control actions, while a cooperative collision avoidance mechanism and action masking ensure both safety and feasibility. Numerical experiments demonstrate that the proposed method can efficiently derive efficiency-enhanced solutions within short computation times.**

***Index Terms*—Virtual coupling, Train group trajectory optimization, Multi-agent deep reinforcement learning**

I. INTRODUCTION

To cope with the increasing freight transport demand, heavy-haul railways characterized by extended train lengths and increased axle loads have significantly enhanced transport capacity [1]. However, these advances have precipitated frequent incidents, including rail/coupler failures and squeeze accidents, accompanied by operational challenges such as intermediate station inaccessibility, prolonged departure intervals, and slow acceleration performance [2]. Existing heavy-haul railway dispatching systems rely on moving block signaling control based on absolute braking distance tracking. To further optimize line transport capacity, implementing train groups with reduced temporal headway has emerged as a more effective strategy for improving operational efficiency [3]. This approach leverages advanced control methodologies to maintain safe, yet minimized, inter-train spacing, thereby maximizing infrastructure utilization while addressing the inherent limitations of conventional heavy-haul operations [4].

Through train-to-train wireless communication and dynamic cooperative control mechanisms, virtual coupling technology allows multiple trains to operate as an integrated convoy. Employing a leader-follower control strategy effectively reduces the relative braking distance (RBD) [5]. In a group, trains typically operate as a cohesive unit within a section, with each train acting as an independent agent. Based on global objectives and its own state, each train dynamically computes a reference speed profile, tracks the state of the leader train, and ensures a dynamically safe relative braking distance [6]. However, the complex state transition mechanisms and real-time cooperative control requirements pose significant challenges for optimizing speed profiles in heavy-haul train groups. Traditional fixed-block signaling systems, designed for static train separation, fail to meet the dual demands of flexibility and safety under virtual coupling. Furthermore, the stochastic disturbances inherent in real-world operations impose heightened robustness requirements on speed profile optimization. The stochastic nature of disturbances may subject individual trains to varying speed constraints, potentially leading to group disintegration and compromised system stability. Therefore, an adaptive control framework capable of coordinating dynamic safety margins with synchronization requirements in uncertain environments becomes particularly essential.

For virtual coupling, Quaglietta et al. [7] developed an innovative train-following model that systematically characterizes operational state transitions in virtually coupled trains. Addressing the demand for tracking interval compression in virtual coupling, Felez et al. [6] pioneered the application of model predictive control to leader-follower train coordination, significantly reducing inter-train headway. Zhang et al. [8] introduced a fixed-time cooperative control method that optimizes multi-train departure strategies with predetermined coupling time constraints. To further enhance multi-train coordination efficiency, Cao et al. [9] developed a hybrid control architecture integrating generalized model prediction with artificial potential field methods, achieving distributed cooperative control of virtually coupled trains. Focusing on heavy-haul train scenarios, Zhang et al. [3] employed dynamic programming to optimize following trajectories, achieving minimal tracking distances while respecting the physical constraints of traction/braking systems. For stochastic disturbance mitigation, Wang et al. [10] innovatively incorporated stochastic initial delays into optimization models, designing real-time cooperative trajectory planning algorithms that balance energy efficiency, punctuality, and safety objectives through dynamic route selection, thereby extending virtual coupling applicability to non-ideal scenarios.

In recent years, reinforcement learning has shown significant potential in optimizing train trajectory and timetable, with its core strength in autonomous exploration and trial-and-error, enabling dynamic decision optimization in complex

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scenarios. Early studies focused on Deep Q-Network (DQN) methods [11], [12] with discrete action spaces. However, since train operation control inherently involves continuous action spaces, scholars introduced the Deep Deterministic Policy Gradient (DDPG) algorithm [13]. As virtual coupling systems demand enhanced multi-train coordination, research has evolved toward collaborative optimization paradigms: Su et al. [14] integrated trajectory prediction with DQN to establish a train platoon safety model. Liu et al. [15] applied multi-agent reinforcement learning (MARL) for distributed acceleration/braking coordination, enhancing group synchronization. Building on this, Liu et al. [16] proposed a Giraph Neural Networks with attention mechanisms to solve the joint timetable and platform rescheduling problem under virtual coupling. Dong et al. [17] further explored deep reinforcement learning for integrating train trajectory optimization and timetable rescheduling under disturbances.

Although existing studies have made breakthroughs in model construction and algorithm design, most research predominantly focuses on the control strategies of following trains and still faces challenges such as unclear mechanisms of multi-train coupled dynamics, cooperative control difficulties in train formations, and complex interactions among multi-source disturbances. Particularly under heavy-haul conditions, inertial effects amplify the complexity of control. Moreover, traditional Multi-Agent Deep Deterministic Policy Gradient (MADDPG) lacks a goal-oriented mechanism, leading to slow convergence. This paper will address the resilience to temporary speed restrictions (TSRs), explore hybrid methodologies integrating data-driven approaches with physical models, and implement reinforcement learning-based online optimization in complex environments. The contributions of this work are summarized as follows:

1) A mixed-integer quadratic programming (MIQP) model that incorporates TSRs disturbances is developed to optimize disturbance-adaptive reference speed curves for each train in the group.
2) A GO-MADDPG algorithm is proposed, which integrates a goal-oriented guidance mechanism to accelerate convergence and a cooperative collision avoidance strategy to ensure operational safety.
3) A flexible decoupling strategy tailored for TSRs scenarios is designed to balance operational efficiency and dynamic stability of the train group.

The remainder of this paper is organized as follows. Section II presents the problem description and model formulation. Section III proposes an MARL algorithm, providing comprehensive details about the algorithmic enhancement strategies. Section IV conducts numerical experiments to validate the efficacy of the proposed approach. Finally, Section V gives the conclusions.

II. Problem Description and Model Formulation

A. Problem Description

Group operation refers to the cyber-physical coupling of distributed heavy-haul trains through computer and network communication technologies, replacing mechanical couplers to form logically integrated train groups operating as unified entities. Throughout their lifecycle, train groups undergo phase-specific state transitions characterized by dynamic reconfiguration capability, enabling flexible coupling/decoupling operations. This operational paradigm introduces multilayered complexity in line capacity utilization compared to conventional train control systems. Particularly under TSRs, the stochastic nature of speed constraints may impose heterogeneous velocity limitations across group formations. Such non-uniform disturbances amplify the risk of dynamic state desynchronization between leading and following units, ultimately compromising the group's coherent motion stability.

In this paper, the optimization of the speed trajectory for leader-follower trains is approached by modeling each train within the group as an independent agent. This method simultaneously optimizes the operational trajectories of both leader and follower trains, ensuring overall operational efficiency and dynamic stability of the train group. The overall time horizon is uniformly discretized into T periods of equal time length. The impact of TSRs is considered, with speed limits beginning or ending at any time or location. As illustrated in Fig. 1, consider a train group composed of three trains (A, B, and C) operating in coordinated formation. To facilitate virtual coupling operation, the coupling relationships among these trains are established at the station prior to departure. The formation initially maintains its virtual coupling state during operation, then gradually transitions to a group operation mode. When encountering TSRs, the train group dynamically decouples into subgroups - with some trains continuing to operate in group formation while others switch back to individual train unit operation.

The following assumptions are made for establishing the trajectory optimization model of train groups:

1) Trains start simultaneously from standstill with initial safety spacing (consistent with station marshalling procedures, ensured by constraint (21)).
2) Train length is explicitly considered (incorporated in safety spacing calculation).

B. The objective Function

The definitions of symbols and variables in the optimization model are summarized in Table I.

The objective function minimizes a weighted sum of train travel time (Z_1), operation cost (Z_2), and stability deviation (Z_3). Z_2 includes energy consumption, calculated as $E_{t_sum} = \sum_{k\in T} F_{t_i}(k)\cdot\triangle l$ (traction work), and air brake cost $E_{c_sum} = \sum_{k\in T} \mu_{c_i}(k)\cdot\triangle t$. To address the disparate magnitude scales of the three objectives, a normalization procedure is implemented to unify their dimensionalities within the optimization framework, thereby eliminating scale dominance effects in the multi-objective decision space. d_{des} (minimum desired spacing) is calculated as $f(v_i, v_{i-1})$ based on RBD. Z is used to denote the objective function, which can be described as follows.

$$\min\ Z = w_1 \cdot Z_1 + w_2 \cdot Z_2 + w_3 \cdot Z_3 \tag{1}$$

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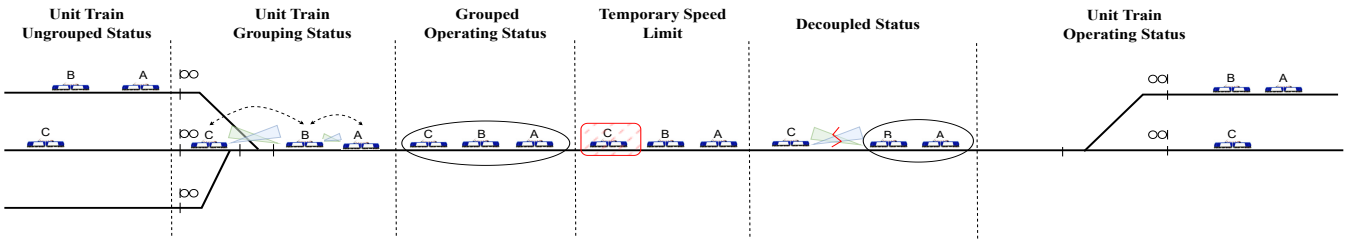


Figure 1. Schematic diagram of status of each stage of train group operation.

Table I
PARAMETERS AND VARIABLES

Notation	Definition
N	Set of trains.
T	Set of section running time.
T_s	Maximum section running time.
$F_{t_{max}}, F_{d_{max}}$	Maximum traction force and brake force.
v_{max}	Maximum permitted speed.
$\triangle t$	Minimal discretisation time.
$\triangle l$	Distance travelled in discretisation time $\triangle t$.
d_{des}	Minimum desired spacing for coupled trains.
M	Train weight.
S	Length of Section.
W_0, W_p	Basic resistance and slope resistance.
g_i	Line slope.
L_l	Train length.
S_{max}	Minimum safe margin.
v_{lim}	Maximum temporary speed limit.
v_{rmin}	Minimum release speed.
t^a	Minimum application time of the air brake.
t^r	Minimum release time of the air brake.
t^s	Start time of temporary speed limit.
t^e	End time of temporary speed limit.
Variable	**Definition**
s_k	Distance at time k.
v_k	Speed at time k.
T^r	Section running time.
F_t	Traction force.
F_d	Brake force.
F_c	Air brake force.
R	Resistance.
μ_t, μ_d, μ_c	0-1 variable, the force is applied at time t.
d_k	Difference in distance between coupled trains.
L_{brake}	Maximum braking distance of the train at time t.
ζ_{i_j}	Train i is located within the speed limit.

$$Z_1 = \sum_{i \in N} \frac{T_i^r}{T_s} \tag{2}$$

$$Z_2 = \sum_{i \in N} \sum_{k \in T} \frac{F_{t_i}(k) \cdot \triangle l}{F_{t_{max}} \cdot S} + \frac{\mu_{c_i}(k) \cdot \triangle t}{T_s/2} \tag{3}$$

$$Z_3 = \sum_{i \in N} \sum_{k \in T} \frac{|v_i(k) - v_{i-1}(k)|}{v_{max}} + |1 - \frac{d_i(k)}{d_{des}}| \tag{4}$$

C. Constrains

$$a_i(k) = \frac{\mu_t F_{t_i}((k) - \mu_d F_{d_i}(k) - \mu_c F_{c_i}(k) - R(s_k, v_i(k))}{M_i} \tag{5}$$

$$s_i(k+1) = s_i(k) + \frac{1}{2}(v_i(k) + v_i(k+1)) \cdot \triangle t \tag{6}$$

$$R(s_k, v_i(k)) = 0.001 \cdot (W_0 + W_p) \cdot M_i g \tag{7}$$

$$W_0 = a + b \cdot v_i(k) + c \cdot v_i(k)^2 \tag{8}$$

$$W_p = g_i \tag{9}$$

$$v_i(k+1) = v_i(k) + a_i(k) \cdot \triangle t \tag{10}$$

$$0 \leq F_{t_i}(v_i(k)) \leq F_{t_{max}} \cdot \mu_{t_i}(k) \tag{11}$$

$$0 \leq F_{d_i}(v_i(k)) \leq F_{d_{max}} \cdot \mu_{d_i}(k) \tag{12}$$

$$\mu_{t_i}(k) + \mu_{d_i}(k) \leq 1, \forall i \in N, k \in T \tag{13}$$

$$\mu_{c_i}(k) \leq \mu_{d_i}(k), \forall i \in N, k \in T \tag{14}$$

$$v_i(0) = 0, v_i(S) = 0, \forall\ i \in N \tag{15}$$

$$v_i(k) \leq v_{max}, \forall\ k \in T \tag{16}$$

$$v_i(k+1) \geq v_{rmin}, if\ \mu_{c_i}(k) - \mu_{c_i}(k+1) = 0 \tag{17}$$

$$\textstyle\sum_{j=1}^{t_a} \mu_{c_i}(j) = t_a, if\ \mu_{c_i}(k+1) - \mu_{c_i}(k) = 1 \tag{18}$$

$$\textstyle\sum_{j=1}^{t_r} \mu_{c_i}(j) = 0, if\ \mu_{c_i}(k) - \mu_{c_i}(k+1) = 1 \tag{19}$$

$$d_i(k) = max(L_{brake,i} - L_{brake,i-1} + L_l + S_{max}, L_l + S_{max}) \tag{20}$$

$$s_i(k) - s_{i-1}(k) \geq d_i(k), \forall i, i-1 \in N, k \in T \tag{21}$$

$$L_{brake,i} = v_i^2/2a_i, \forall i \in N \tag{22}$$

$$(1 - \zeta_{i_j}) \cdot v_{max} \geq v_i(j) - v_{lim}, \forall j \in [t^s, t^e], i \in N \tag{23}$$

Constraints (5)-(10) govern train dynamics through temporally discretized motion equations derived from Newtonian mechanics. Constraints (11)-(12) bound the operational ranges of traction and braking forces within physical actuator limits. Constraint (13) enforces mutual exclusivity between traction and braking force applications, while constraint (14) mandates concurrent activation of electrodynamic braking during air brake engagement. Boundary conditions are specified in constraint (15), defining initial and terminal operational states. Constraint (16) imposes line speed restrictions to ensure safety. Safety-critical braking logic is embedded in constraints (17)-(19): constraint (17) prevents velocity from dropping below the minimum release threshold during air brake pressure decay, with constraints (18)-(19) regulating air brake application/release timing parameters. Group operation requirements are formalized in constraints (20)-(22), which enforce inter-train spacing requirements through safety thresholds based on relative velocity and braking performance. Finally, constraint (23) implements TSRs compliance by enforcing speed reduction protocols for affected trains.

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III. Goal-Oriented MADDPG Algorithm

The essence of group operation curve optimization is to determine optimal strategies for each train in the group. Based on the division of inter-station running time, the group operation process between stations can be regarded as a multi-stage decision-making problem. To solve the optimization model constructed in Section II while considering coordinated curve optimization for all trains, a MARL approach is proposed. Since speed curve optimization is a sparse goal-oriented problem, self-supervised learning is used to guide the reinforcement learning training process. Additionally, expert experience is incorporated via a cooperative collision avoidance mechanism to improve optimization effectiveness. The corresponding algorithm architecture is illustrated in Fig. 2. The methodology is developed in three steps: First, the train group operation problem is formulated as a multi-stage decision-making process. Subsequently, relevant elements of group operation are mapped to the RL framework for systematic training. Finally, improvements to the MADDPG algorithm tailored for group operation scenarios are elaborated.

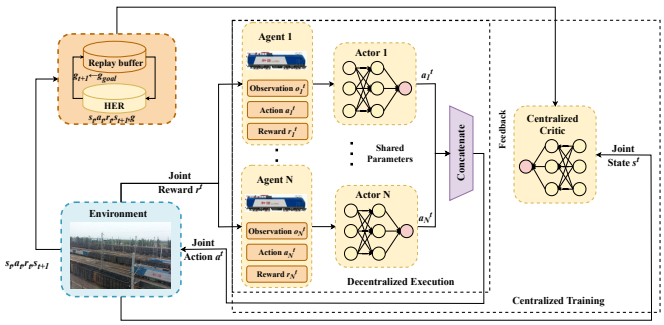


Figure 2. The framework of goal oriented MADDPG.

A. Formulation of Markov decision process

The operation of train groups follows a continuous cycle of receiving operational information and executing control commands. Each train starts with state variables (position, velocity, acceleration, and temporal parameters). Upon receiving commands from the Automatic Train Operation (ATO) system and preceding trains, the group operates for a designated interval, then updates its state. This iteration continues with revised instructions. When disturbances occur, the group may either decouple or adaptively maintain coordinated movement, ultimately converging to target states. Analyzing state information and control commands throughout this process enables deriving optimized speed profiles for the entire group, thus transforming the speed curve optimization into a multi-step decision-making framework.

a) Environment: The operational environment is systematically defined by integrating three critical components: train dynamics, ATP (Automatic Train Protection) speed protection profiles, and line infrastructure conditions.

b) State: To satisfy the optimization objectives outlined in Section 2, this section selects group trains 1, 2, and 3 as the state variables, which are normalized as shown in the equations below. The initial and target states are defined as follows.

$$S_k^n = \left\{ s_n^k, v_n^k, a_n^k, t_n^k, v_{n_k}^{lim}, E_{t_{n_k}}^{sum}, E_{c_{n_k}}^{sum}, s_{n_k}^g, ac_n^k \right\}, \quad \forall\, n = 1,2,3; k = 1,2,\ldots,T \tag{24}$$

where S_k^n represents the state of agent n at step k. $E_{t_{n_k}}^{sum}, E_{c_{n_k}}^{sum}, s_{n_k}^g, ac_n^k$ denote the accumulated energy consumption, the accumulated cost of the air brake, the target position, and the implementation of the air brake of agent n at step k.

c) Action: The agent serves as the controller for individual trains within the group, regulating their actual output forces. The action is defined as a continuous value between -1 and 1 ($-1 \leq a \leq 1$), representing the normalized ratio of the actual output force to the maximum force. When the value exceeds 0, a tractive force is generated by the train; conversely, a braking force is applied when the value is less than 0. Air brake activation requires simultaneous electrodynamic braking (constraint (14)), with minimum application time to avoid system damage.

d) Reward: Derived from the global state, the reward function reflects expected training outcomes and guides learning. Aligned with the objective functions in Section 2, it uses negative values of Z_1, Z_2, Z_3 to match the minimization goal. A negative value Θ serves as a penalty when agents violate constraints (e.g., maximum speed, minimum speed during air brake release, minimum tracking interval, or temporary speed limits). Conversely, a reward is granted for successful destination arrival without constraint violations.

$$R = \begin{cases} \Theta, & if\ (16),\ (17),\ (21)\ or\ (23)\ are\ violated \\ -(w_1 Z_1 + w_2 Z_2 + w_3 Z_3), & else \end{cases} \tag{25}$$

B. Improved goal-oriented MADDPG

In most group train speed profile optimization frameworks, rewards are primarily concentrated on terminal objectives, leading to sparse reward signals. With few intermediate rewards, purely exploratory strategies struggle to identify optimized trajectories that meet operational metrics. To enhance the applicability of RL in multi-train speed profile optimization, this study integrates goal-oriented guidance and a cooperative collision avoidance mechanism to refine the learning algorithm.

• Goal-oriented guidance

During interaction with the environment, when an agent fails to achieve its original goal, it can replay experiences by hypothesizing alternative goalsconverting ”failed” experiences into valuable learning signals. This mechanism innovatively expands representational space of traditional RL: system states now explicitly encode target task information alongside environmental observations. The resulting state-goal-action triad (Eq. 26) provides a multi-dimensional decision basis, enabling

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dynamic policy optimization for diverse objectives. By maximizing expected rewards, agents gradually map composite state spaces to optimal action policies, significantly boosting learning efficiency and strategic adaptability in complex environments.

$$< s_t, g, a_t > \rightarrow Replay\ Buffer \tag{26}$$

• Collaborative collision avoidance mechanism

During a speed curve search, agents often violate speed limits or maintain insufficient leader-follower spacing (below the minimum safety margin). Solely relying on negative rewards fails to guarantee operational safety. To address this, a dual-layer safety strategy is proposed: (1) After each state update, dynamic calculation of maximum/minimum safety protection curves defines the range of executable actions. (2) Action masking bounds action selection, ensuring subsequent maneuvers are safe and feasible. This hierarchical constraint enables proactive risk mitigation while preserving algorithmic flexibility.

C. Training process

The learning procedure of the proposed goal-oriented MADDPG framework incorporates temporally correlated noise into the policy update mechanism to enhance exploration diversity. The intensity of this noise is progressively attenuated during training, enabling agents to asymptotically converge toward optimal policies. Furthermore, to mitigate overfitting risks, a regularization term with weight decay is applied to the critic network, thereby stabilizing the learning dynamics while preserving generalization capabilities.

IV. Numerical Experiment

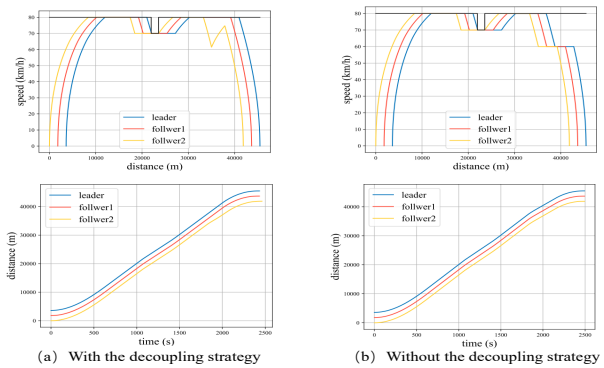
In this section, we applied the real-world data of HXD1-type electric locomotives from the Shenchinan-Longgong section of the Shuo-Huang Heavy-Haul Railway to validate the proposed method through simulation experiments. The parameters of the train, railway line, and algorithm are presented in Table II, with a safety protection distance set at 100 m. The simulation was implemented using Python 3.9 with PyTorch on a Windows 11 workstation equipped with an Intel Core i5-13500H CPU and 32 GB RAM. The experiment comprised two comparative analyses: Firstly, the flexible decoupling strategy was compared with conventional non-decoupling approaches under speed-restricted scenarios to demonstrate efficiency improvements in train formation operations. Secondly, the enhanced algorithm was evaluated against baseline strategies to verify the effectiveness of proposed methodological improvements. A detailed exposition of the experimental process and results follows below.

A. Comparison of flexible decoupling strategy

In scenarios where nonuniform speed restrictions disrupt train group operations, traditional emergency responses prioritize safety by mandating uniform speed reductions for all trains. In contrast, the proposed flexible decoupling strategy enables targeted adjustment only for affected trains, thereby preserving overall transportation efficiency. To validate this approach, we compared optimized speed trajectories under varying speed-restriction scenarios using both the flexible decoupling and conventional methods. As shown in Fig. 3, Fig. 3(a) demonstrates that the flexible decoupling strategy only affects the tail train while achieving a shorter total travel time. Detailed objective function values in Table III reveal a trade-off between operational efficiency and energy consumption: the "-" scenario (Index 1) in Table III refers to the baseline with no TSRs, where all trains maintain virtual coupling throughout the journey. Compared to the non-decoupling strategy, the decoupling approach improves operational efficiency by 2.03% but incurs a 1. 21% increase in energy consumption. This slight energy rise stems from affected trains adjusting traction to adapt to speed limits and resume efficient operation, ensuring overall efficiency. In particular, the collision avoidance mechanism maintains the dynamic stability of the coupled formation. These results confirm the flexible dispatch strategy enhances speed trajectory optimization, balancing safety and efficiency.

Table II
THE PARAMETERS IN ALGORITHM 1

Parameter	Value	Parameter	Value
Train length	1567m	Buffer size	10^8
Agent initial position	[3600,1800,0]	Learning rate	$1e^{-4}, 1e^{-3}$
Agent terminal position	[45465, 43665, 41865]	Discount rate	0.95
Time Step $\triangle t$	$1s$	τ	0.005
Maximum training episode	1000	$\triangle sm$	100
Batch size	128	$v_{max}(km/h)$	80
Layers of two neural networks	3,3	Weight coefficient	0.6, 0.3, 0.1



(a) With the decoupling strategy (b) Without the decoupling strategy

Figure 3. The trajectories with and without the decoupling strategy under the temporary speed restriction.

B. Comparison of different approaches

To validate the performance of the proposed algorithm, this section compares the MADDPG, goal-oriented MADDPG methods. The goal-oriented MADDPG framework facilitates guided exploration of agents toward predefined objectives, thus accelerating convergence of the reward function during training. In contrast, MADDPG implementations lacking explicit

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Table III
THE RESULTS OF DIFFERENT DISPATCH STRATEGIES.

Index	Scope of TSR	Strategy	Objective Value	Total running time (s)	Total operation cost		Total dynamic stability deviation	
					Energy consumption (kWh)	Air brake application (s)	Speed deviation(km/h)	Distance deviation(m)
1	-	-	1.9194	7297	9989.34	3	0.33	646759.66
2	<1800-2400,25000-35000>	Decoupling	1.8964	7328	10111.92	3	0.26	562720.15
		Non-Decoupling	1.9499	7480	9813.22	3	0.28	662987.12
3	<1500-2200,30000-40000>	Decoupling	1.9196	7756	9805.51	3	0.21	398733.56
		Non-Decoupling	2.0120	7777	9627.30	3	0.28	689327.65

goal-oriented mechanisms often exhibit unstable or prolonged convergence behavior due to uncoordinated exploration in multi-agent environments. A comparison of the convergence curves of the reward function is illustrated in Fig 4.

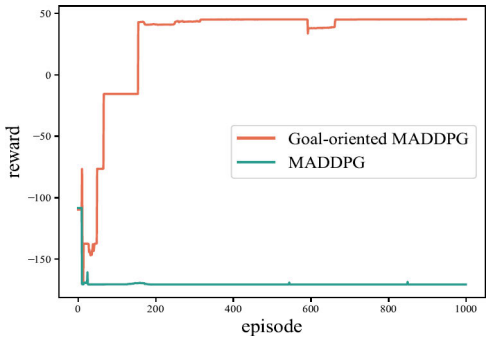


Figure 4. Comparison of reward curves for different methods.

V. CONCLUSIONS

This study addresses the trajectory optimization problem for heavy-haul train group operation under emergent events. By analyzing the collaborative operation of train groups under different decoupling strategies, practical guidance is provided for coordinating train groups across various stages between stations during emergencies. An MIQP model is constructed for the collaborative optimization of heavy-haul train groups. An improved MARL algorithm is designed to derive driving curves for individual trains. The algorithm incorporates a self-supervised experience replay mechanism tailored to real-world heavy-haul operation characteristics, as well as a collaborative collision avoidance mechanism to ensure safe driving. Simulation experiments involving three heavy-haul trains demonstrate the effectiveness and feasibility of the proposed method. Future research will focus on the integrated rescheduling of timetables and collaborative optimization of speed curves for train groups, aiming to eliminate the hierarchical architecture between these two aspects and enhance the overall operational efficiency of heavy-haul railways.

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